SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

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**Solution Architecture**

**Date:** 15 March 2026

**Team ID:** *(xxxxx)*

**Project Name:** Human Development Index (HDI) Prediction Using Machine Learning

**Maximum Marks:** 5 Marks

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**Solution Architecture Diagram Content**

**Presentation / Client Layer**

**Technologies:**

- HTML5
- CSS3
- Bootstrap
- JavaScript

**Purpose:**
Provides a user-friendly web interface where users enter the following details:

- Life Expectancy
- Mean Years of Schooling
- Expected Years of Schooling
- GNI Per Capita

Displays the predicted HDI category.

⬇ ⬆

**API Gateway / Web Server**

**Technology:**

- Flask (Python Web Framework)

**Purpose:**

- Receives user requests.

- Sends input data to the machine learning model.
- Returns prediction results to the web page.

⬇ ⬆

**Auth Service**

**Technology:**

- Flask Session (Optional)

**Purpose:**

- Basic user authentication (if implemented).
- Manages user sessions securely.

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**Core Logic Service**

**Technologies:**

- Python
- Scikit-learn
- Pandas
- NumPy
- Joblib

**Purpose:**

- Reads user input.
- Performs data preprocessing.
- Loads the trained machine learning model (model.pkl).
- Predicts the HDI category.
- Sends prediction results back to the Flask application.

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**External API Integrations**

**Technology:**

- No external APIs used.

**Purpose:**
The project works completely offline using a local dataset and trained machine learning model.

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**Data / Storage Layer**

**Technologies:**

- CSV Dataset
- Joblib (model.pkl)

**Purpose:**
Stores:

- HDI dataset
- Trained machine learning model
- Input data for prediction

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**Component Description Table**

| Component Name | Description / Role in Architecture | Technologies Used |
|---|---|---|
| Presentation Layer | Collects user input and displays the predicted HDI category through a web interface. | HTML5, CSS3, Bootstrap, JavaScript |
| API Gateway | Handles HTTP requests, processes form data, and connects the frontend with the machine learning model. | Flask (Python) |
| Core Logic Service | Performs data preprocessing, loads the trained model, predicts the HDI category, and returns the prediction result. | Python, Scikit-learn, Pandas, NumPy, Joblib |
| Database / Storage | Stores the HDI dataset and the trained model for prediction. | CSV File, Joblib (model.pkl) |

## VS Code Project Structure

```
HDI-Prediction/
│
├── dataset/
│   └── hdi_dataset.csv
│
├── models/
│   └── model.pkl
│
├── static/
│   ├── style.css
│   └── script.js
│
├── templates/
│   └── index.html
│
├── app.py
├── train_model.py
├── requirements.txt
├── README.md
└── .gitignore
```

## Team Members

| Team Member | Role | Responsibility |
| --- | --- | --- |

| Team Member | Role | Responsibility |
|---|---|---|
| G Naveen | Team Lead | Project planning, GitHub management, and project integration |
| Chandana Busireddy | Member | Dataset collection, preprocessing, and documentation |
| Gandham Joshna | Member | Data visualization and exploratory data analysis |
| Nandineni Gowthami | Member | Flask web application development and testing |
| Shaik Talha | Member | Machine learning model training, report preparation, GitHub updates, and final presentation |

This content is tailored for your **HDI Prediction** project using **Python, Flask, Scikit-learn, and VS Code**, and is ready to be copied into your **Solution Architecture** reference document. Replace the placeholder **Team ID** with your actual team number before submission.